

Introduction to AWS Cloud Computing

Amazon Web Services (AWS) is a cloud computing platform that provides on-demand computing, storage, databases, networking, security, and application services.

Amazon EC2 is a compute service that provides resizable virtual servers called instances. EC2 can be used to run web applications, APIs, background jobs, and other workloads.

Amazon S3 is an object storage service. It can store files such as documents, images, videos, backups, and application assets. S3 organizes objects inside buckets.

Amazon DynamoDB is a managed NoSQL database designed for applications that need fast and predictable performance. It stores data as items inside tables.

AWS Lambda is a serverless compute service. Developers can run code without managing servers. Lambda functions are commonly triggered by API requests, events, or scheduled jobs.

Amazon API Gateway is a service for creating and managing APIs. It can receive HTTP requests and invoke backend services such as Lambda functions.

Amazon Bedrock provides access to foundation models through managed APIs. Applications can use these models for tasks such as question answering, summarization, and text generation.

Retrieval-Augmented Generation (RAG) combines information retrieval with a language model. A RAG system first retrieves relevant document content and then provides that content to the language model so the generated answer can be grounded in the source material.